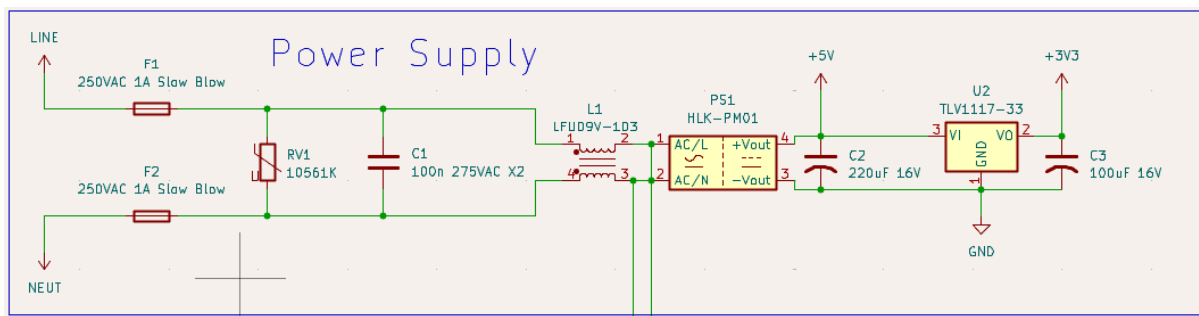


3D Printer Filament Dryer & Print Annealer

Electronics Design Report

Low Voltage PSU Section



HLK-PM1 : 5V 600mA galvanically isolated power supply.

Cause of selection: Small footprint, cost effective, isolated (safety), ease of design.

1117-33 : 3.3V LDO linear regulator.

Cause of selection: To ensure the cleanest signal at the MCU power rails.

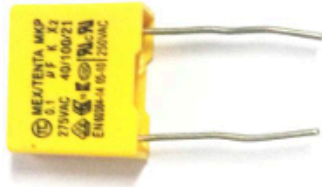
The combination of these two devices allows for the design flexibility and effectively makes the design future proof, by enabling the use of both 5V and 3.3V devices, with main focus on 3.3V.

The PSU section is capable of supplying the peak current consumption of the ESP32 and the other peripherals without brownout.

A MOV used to protect against HV spikes. A common mode choke coil and a suppression capacitor used to suppress EMI. Two fuses were used with ground fault in mind (at the event of reverse connection of the line-neutral).

The HV stage components are selected by the manufacturer recommendation.

8.1 Input parts

Component number / recommended device	Functions	Recommended value
Fuse	Protect the circuit from damage when the module is working wrong	1A/250Vac, Slow fuse
Varistor	The cumulative surge is to protect the module from damage	10D561K
C1/Safety capacitance	Filtering, safety protection (EMC certification)	0.1uF/275Vac
L1/Common-mode inductance	EMI filtering	Sensible value 10-15mH, current 70-500mA
		
Safety capacitance		Common-mode inductance

Notes:

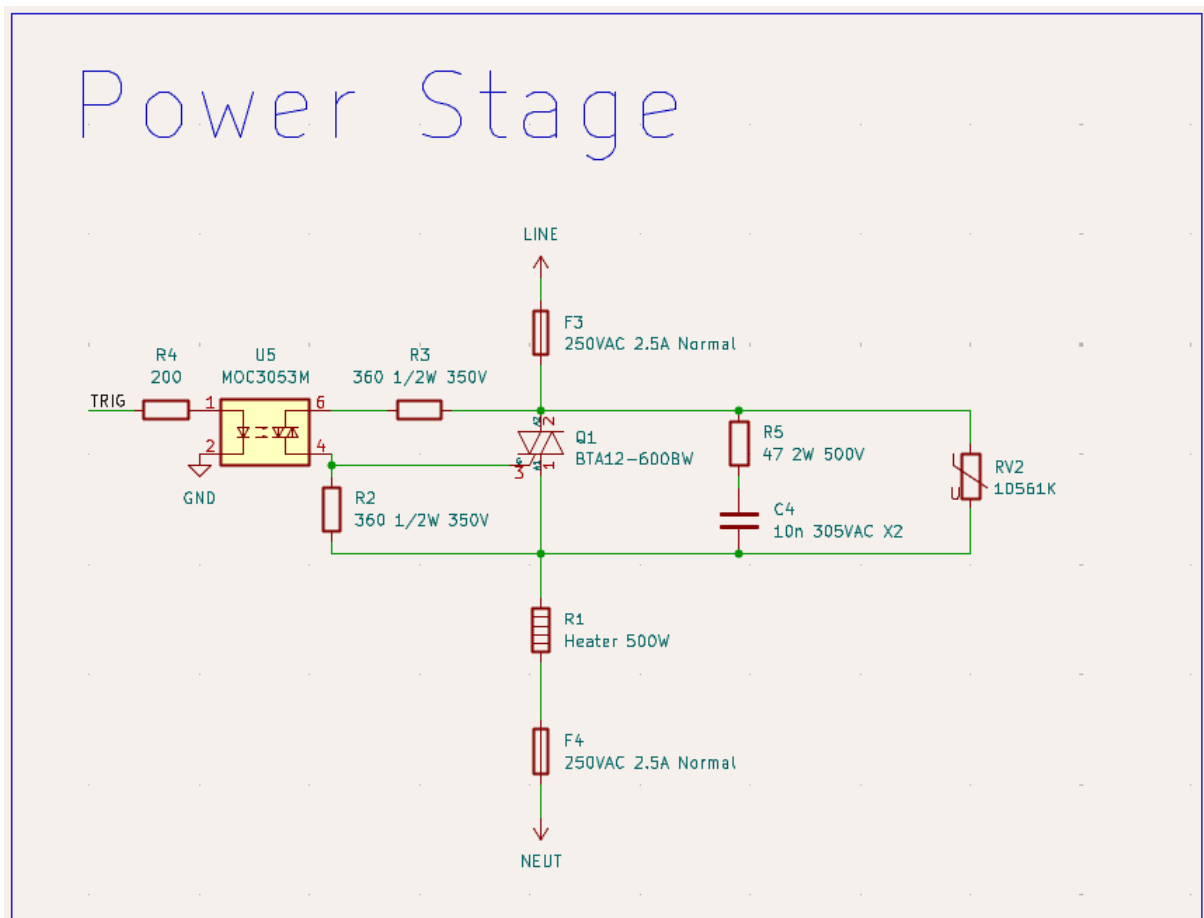
- Fuse and varistor are basic protective circuits (must be connected).
- If you need to pass the authentication/certification, the Safety capacitance and common-mode inductance could not be omitted.
-

The 5V bulk capacitor is also selected by manufacturer recommendation. The higher capacitance value is selected to minimize ripple.

8.2 Output parts

Component number / recommended device	Functions	Recommended value
C2/filter capacitor	output ripple can be controlled in 30mV after adding this capacitor	Aluminium electrolytic capacitance, capacity 100-220 UF, voltage reduction greater than 75%
RL/Load	Load	

Triac / Power Section



For the heating element, a 220V PTC is chosen to decrease the current for the same power. Effectively cutting down components and PCB cost.

The triac control architecture is simpler and a lot cheaper than a half bridge of the same power, yet offers the same precision control capability.

The circuit is designed to control a 500W PTC load safely up to 55°C ambient temperature.

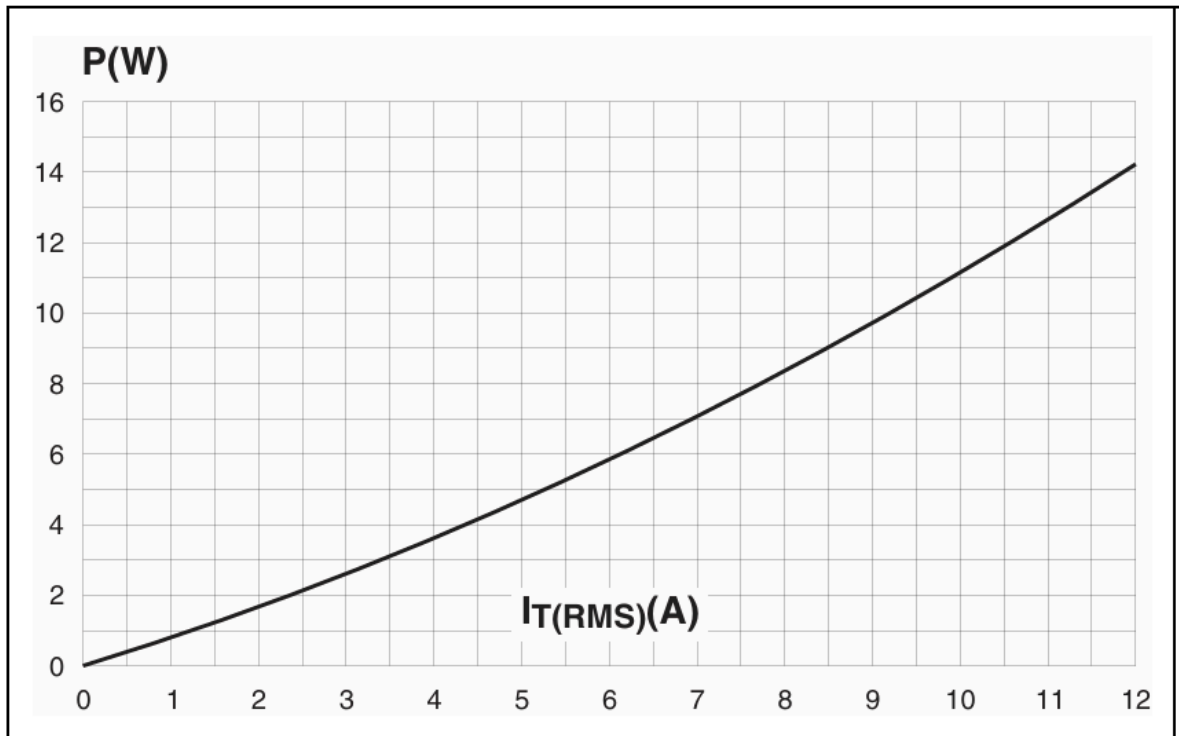
The RMS current consumption of a 500W PTC is $\frac{500W}{220V} = 2.27A$, so the triac must be able to handle at least double this current.

The BTA with W suffix series triacs are good for this job because of the following reasons:

- Isolated tab: safe for heatsinking, allows for mounting on machine chassis)
- Snubberless: designed to handle inductive loads without even needing a snubber
- Has 6 to 16 amp rated models
- Has 600V rated models

The cheapest snubberless BTA series triac at the local supplier (özdisan) was BTA12-600BW.

Power loss and heatsinking calculations of the triac

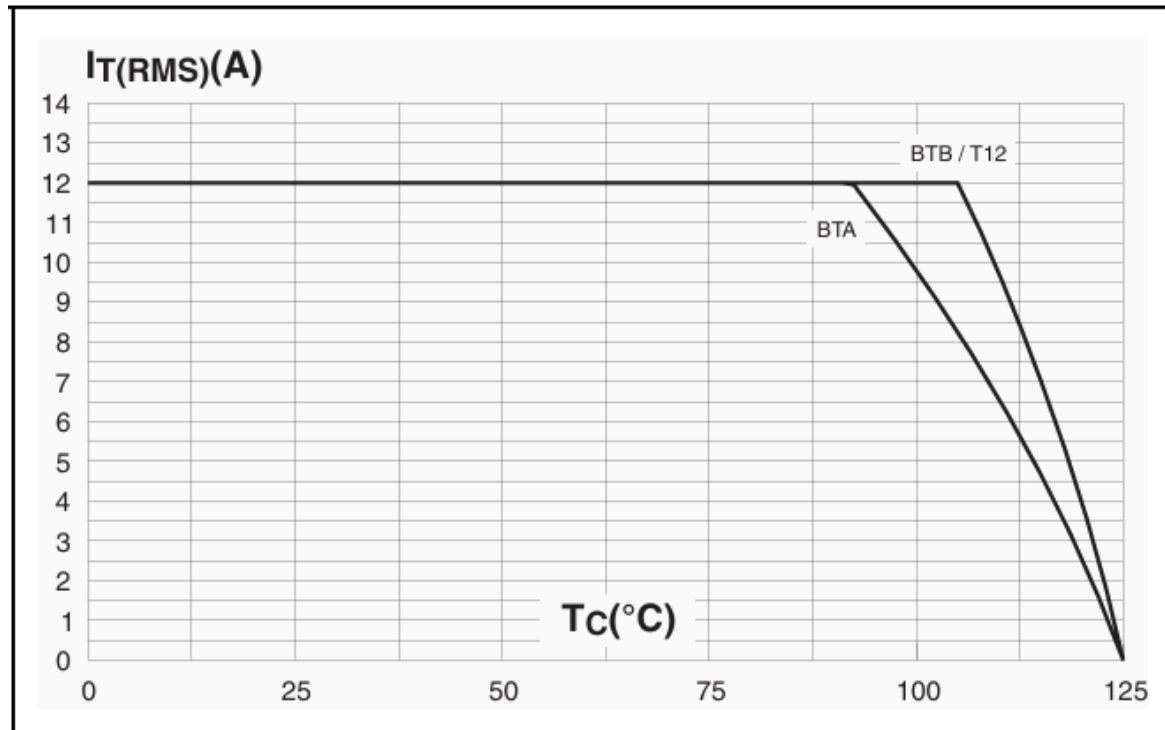
Figure 1. Maximum power dissipation versus RMS on-state current (full cycle)

$$P_d \simeq 2W @ 2.27A \text{ RMS}$$

$$\text{Thermal resistances: } R_{jc} = 2.3^{\circ}C/W \quad R_{paste} \approx 0.5^{\circ}C/W$$

The max T_j is determined from the graph below:

Figure 2. RMS on-state current versus case temperature (full cycle)



As shown in the graph, the case temperature must stay under 90°C for reliable operation, and as a rule of thumb: the junction temperature should be $\leq 75^\circ\text{C}$ to ensure a long service life.

For the max ambient temperature of 55°C, we have a margin of 20 degrees of temperature rise. The low 20K temperature cycling also ensures a long service life by minimizing thermal stress on the silicon.

Needed junction to ambient thermal resistance is: $\frac{20^\circ\text{C}}{2.27\text{W}} = 8.8^\circ\text{C/W}$

So the heatsink R_{th} must be: $8.8 - R_{jc} - R_{paste} = 6^\circ\text{C/W}$

Fuse Selection

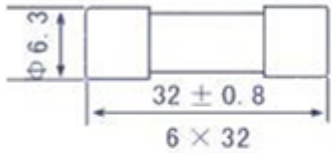
The operating point is designed to be 500W, but there might be surges higher than that. We first need to calculate the maximum safe operation point.

From the latest graph, we can see that the triac must be at $\leq 90^\circ\text{C}$ to operate without degrading. This leaves us with $90 - 55 = 35^\circ\text{C}$ temperature rise margin.

With the calculated $R_{\theta j-a}$ of 8.8°C/W , the max power dissipation is $\frac{35^\circ\text{C}}{8.8} \approx 4\text{W}$

As seen from the first graph, the corresponding RMS current of 4W dissipation is 4.3A

$$\text{Max Load} = 220\text{V} \times 4.3\text{A} = 946\text{W}$$

		BLOWING CHARACTERISTIC TEST		Load	Blowing Time
				150%	min.1hr
				210%	max 30min
				275%	min 50ms max 2s
				400%	min 10ms max 300ms
BLOWING CHARACTERISTIC TEST	Load	Blowing Time	1000%	max 20s	
	210%	< 30min			
	400%	10ms-300ms			

For 2.27A nominal operation point, a 2.5A fast acting fuse is okay. At max load condition, the overload on the fuse is 190%. So the fuse will blow between 30min and 1h. This behavior is good as the max load is a calculated, safe operation point. If the current rises up to 6.8A, the fuse will blow instantly and protect the circuit. Since the BTA12 easily handles 6.8A, the limiting factor is thermal. With this fast acting fuse, the circuit will be protected before it gets dangerously hot in the event of an overload.

Triggering the Triac

An SCR optocoupler is used to isolate the control circuit from the mains, and to drive the power triac with enough gate current. The optocoupler must be non-zero-switching to be able to implement phase angle control.

The MOC3053 chip was found suitable for this job. MOC series optocouplers are widely used and easily procurable. The MOC3053 has a sensitive LED which triggers at 6mA. The peak current rating for the output is 1A, which is highly sufficient for the BTA12-600BW which needs 50mA gate current. The manufacturer recommends running the LED with at least 150% more current to accommodate the degrading of the LED. The forward voltage drop of the LED is specified as 1.18V typ and 1.50V max. The resistor calculation:

$$\frac{3.3-1.18}{9mA} = 235 \quad \frac{3.3-1.5}{9mA} = 200$$

So an 200R resistor will provide the recommended 150% more current (9mA) at worst case, and slightly more current (10.6mA) for the typical scenario. 1/16W power rating provides

$$\sim 64\% \text{ margin. } \frac{(3.3-1.18)^2}{200} = 22.47mW$$

The peak current of the output of the MOC3053 is 1A. Since the peak voltage of the 220V mains is 311V:

$$\frac{311V}{1A} = 311\Omega$$

So we need to place a gate current limiting resistor (R3) of at least 311 ohms. A 360 ohms resistor provides a ~15% margin from the peak current, so is selected.

For a max trigger pulse of 2ms @ 311Vp, the power loss on the resistor is:

$$\frac{220^2}{360\Omega} \times \frac{2}{1000} = 269mW$$

So a 1/2W rated resistor is sufficient. A wirewound resistor is selected because of the robustness.

Additionally, the resistor must be rated for at least 350V.

The same resistor is also used for the gate suppression. This resistor (R2) improves the dV/dt immunity and prevents self or spurious triggering of the triac.

EMI Suppression

Despite the fact that the chosen triac is already a snubberless triac, we must add an RV snubber and a MOV to further ensure the compatibility and safety of the device.

The snubber network is needed to be measured and tested by nature, but the MOC3053 datasheet recommends some values for us.

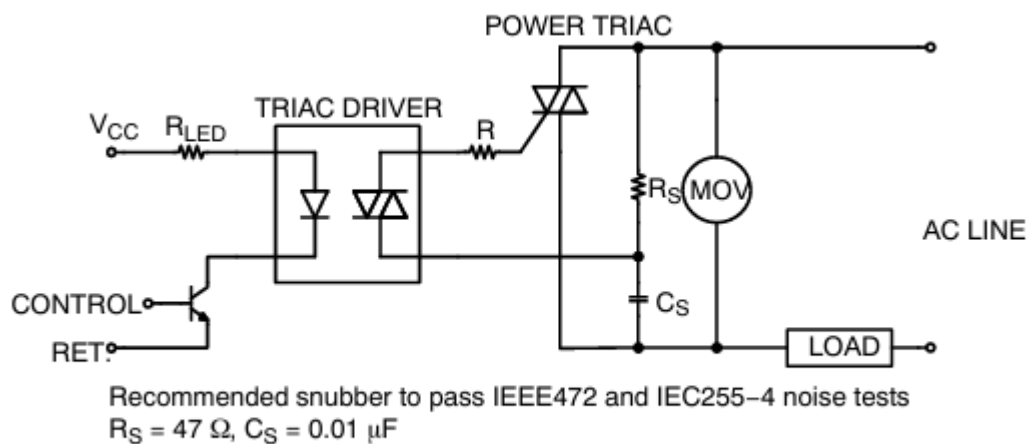


Figure 9. Triac Driver Circuit for Extremely Noisy Environments

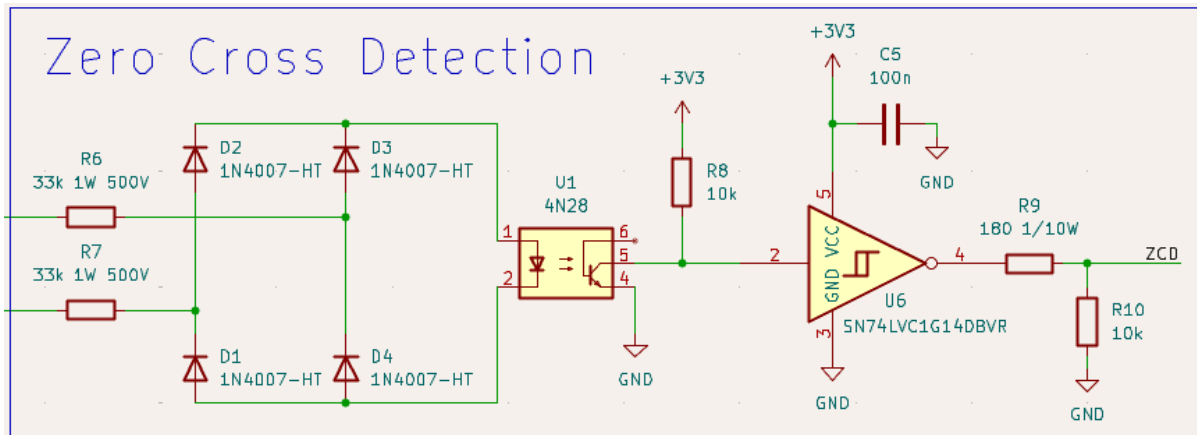
So the recommended snubber values are selected.

The other selection criterion are:

- For the capacitor: Must be X2 rated, must be rated for 305VAC to withstand HV spikes.
- For the resistor: Must be 500V rated, wirewound to ensure robustness. 2W power rating will provide more than enough safety.

For the MOV, the same component is used as in the LV PSU section.

Zero-Cross Detection



The zero cross detection is mandatory to control the triac because we chose a non-zero-crossing triac driver.

So the MCU must be able to detect the zero-crossings of the mains and fire the triac accordingly.

We need an isolated, safe, and cheap way to interface the MCU with the mains.

This zero crossing detection circuit generates a pulse (inverted, active low) at each zero crossing, with a calculable delay.

The architecture is simple: a full wave rectifier using the good old 1N4007 diodes, an optocoupler to isolate the signal, and a schmitt trigger to further refine the signal for the MCU.

The R6 and R7 resistors are needed to protect the LED of the optocoupler.

The two 33k resistors in series results an I_p of $\frac{311V}{66k} = 4.71mA$ which is safe and enough for the 4N28 optocoupler to generate a small pulse. The resistors must be rated for at least 110Vrms (mains/2 because they are in series).

The power loss is $110V \times 4.71mA = 518 mW$

1W rated and wirewound resistors are selected to ensure robustness and long life.

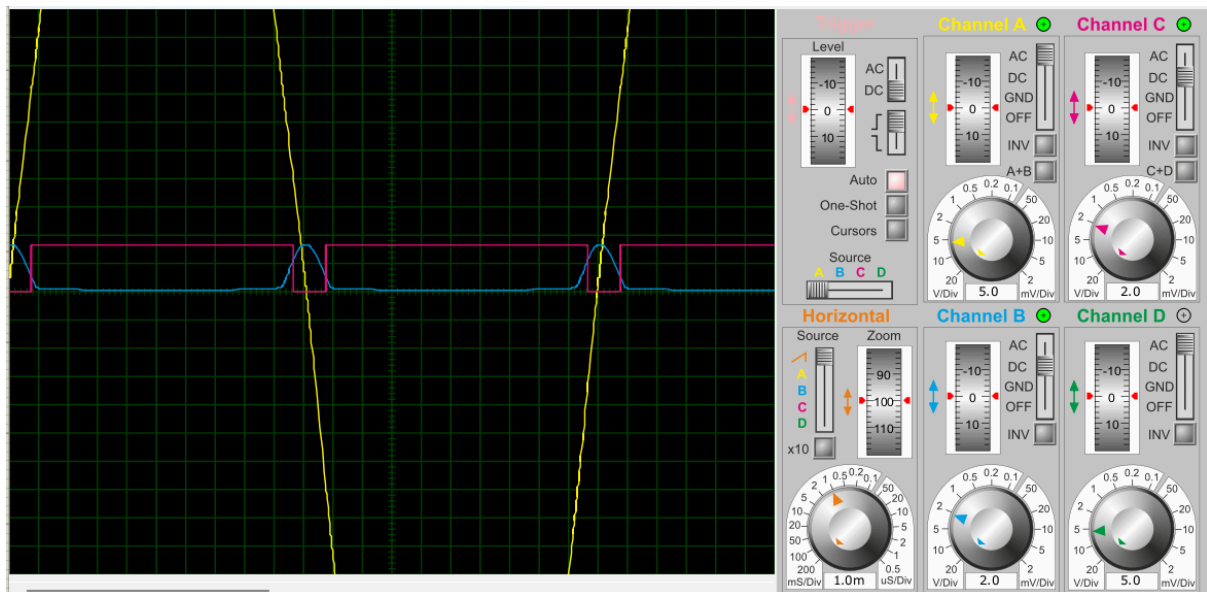
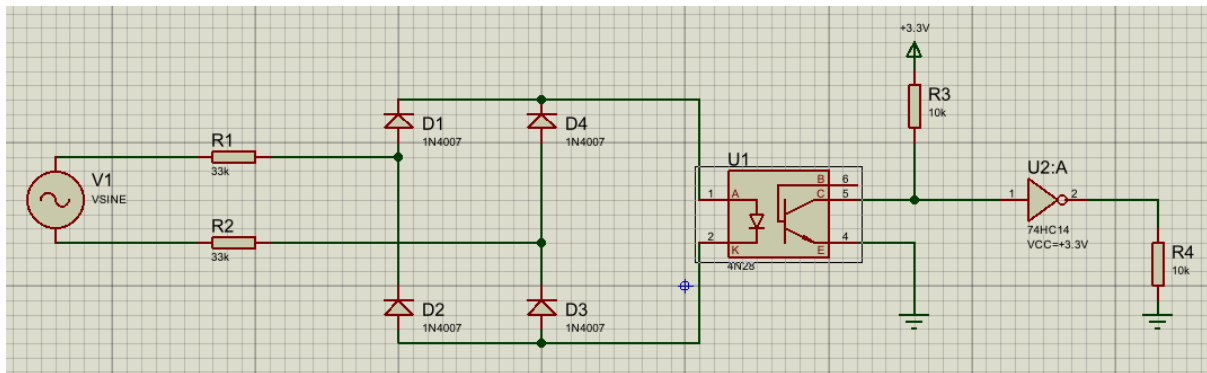
The output is pulled-up with a 10k resistor, so, except the zero crossing event, the transistor stays conductive and drops the output voltage. The schmitt trigger used is a CMOS chip and has high input impedance, so there is no need for an additional resistor to connect the output of the optocoupler.

The schmitt trigger is decoupled with a 100n capacitor.

At the output, a 180 ohms resistor is used to limit the current to ~18mA to both protect the schmitt trigger and the MCU GPIO. A 10k pulldown is added to ensure reliable triggering.

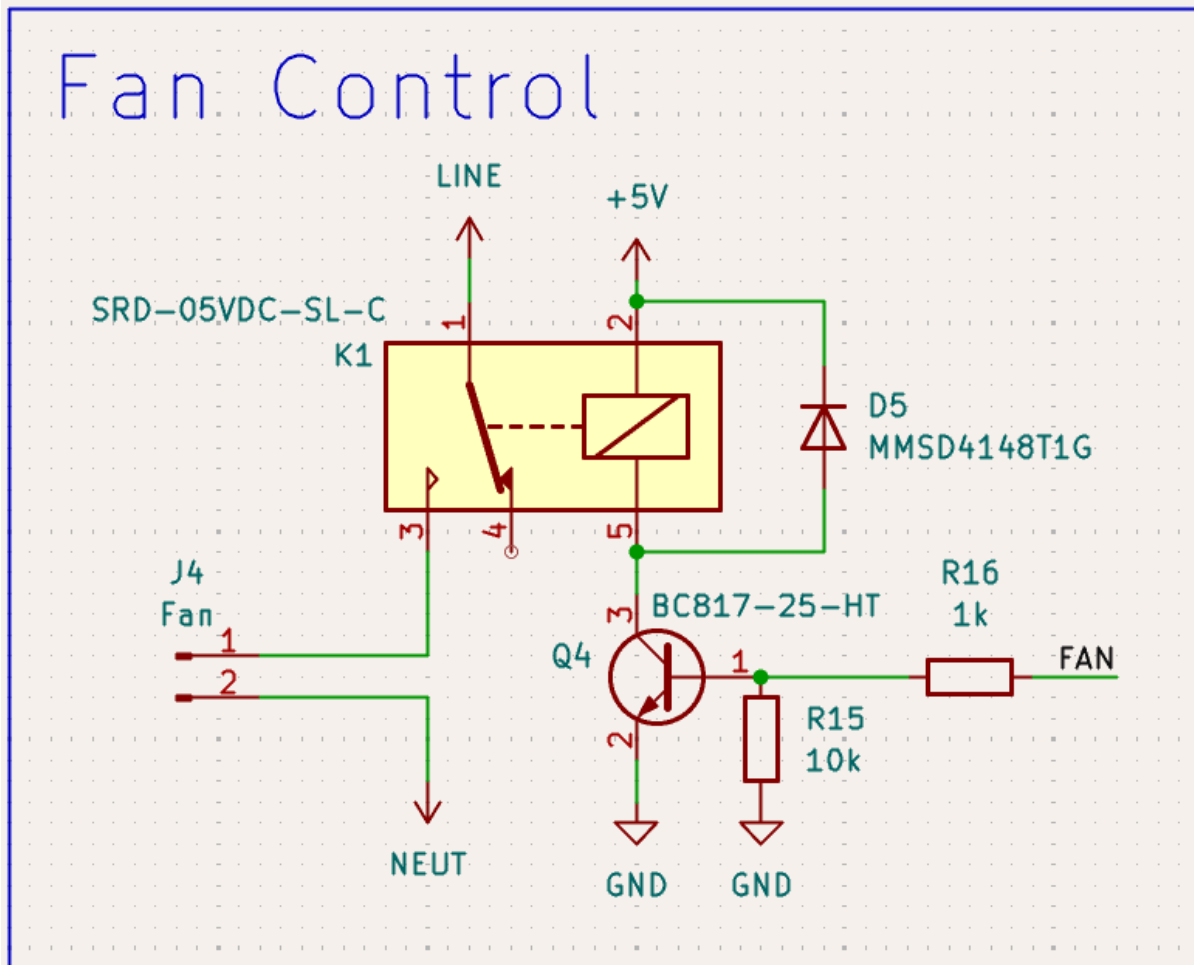
The circuit is tested and the signal is verified using proteus.

ZCD Proteus Simulation



Mains, Optocoupler output, Schmitt trigger output

Fan Control



Relay drive BJT: BC817-25-HT

Cause of selection: 500mA rating, high hFE (up to 390), and cost-effective for small-signal switching.

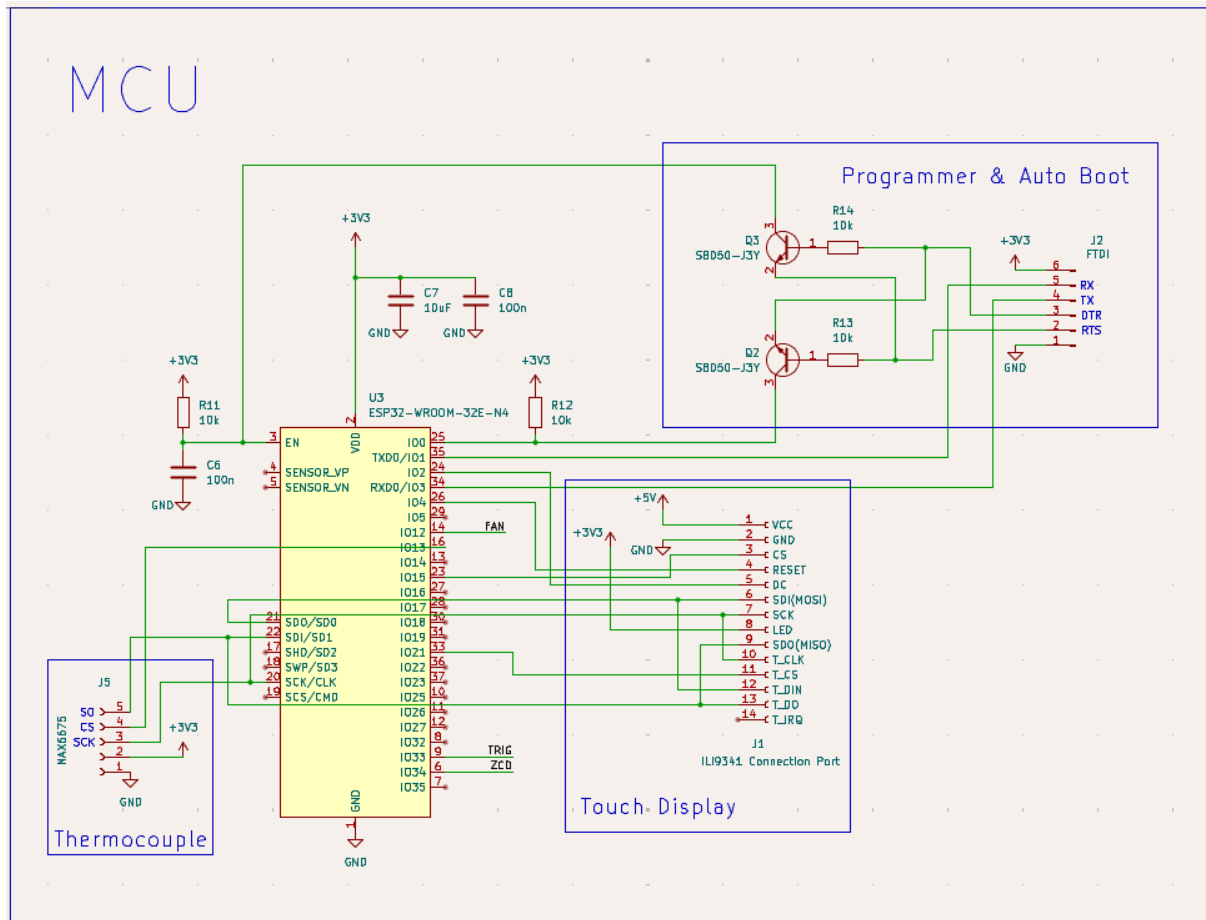
The 70 Ohm relay coil (SRD-05VDC-SL-C) draws 71.4mA at 5V. The 3.3V control signal drives a divider (R16=1k, R15=10k), yielding $V_{th} = 3.3V \cdot (10k/11k) = 3.0V$ and $R_{th} = 909$ Ohms. With V_{be} approx 0.7V, $I_b = (3.0V - 0.7V) / 909 \text{ Ohms} = 2.53mA$. This provides a forced beta of ~28 for hard saturation.

Dissipation is the sum of collector and base losses: $P_d = (71.4mA \cdot 0.1V) + (2.53mA \cdot 0.7V) = 8.91mW$.

Assuming a conservative R_{thja} of 500°C/W at 55°C max ambient: $T_j = 55^\circ C + (0.00891W \cdot 500^\circ C/W) = 59.46^\circ C$

An MM5D4148T1G diode is used for flyback protection. A 10k pulldown is placed to prevent spurious triggering during power-up.

MCU Section



The selected MCU is ESP32-WROOM-32 module. This module has necessary components such as crystal, antenna etc. and simplifies the design process.

The necessary decouplings are done.

An auto boot circuit is implemented for easy programming.

An FTDI port is created with 1x6 2.54mm pin headers.

A touch display is chosen for better UI/UX.

The selected module is ILI9341 TFT with XPT2046 touch controller.

The SPI connections for both the display and touch controller are done.

A 1x14 2.54mm pin header port for connection. Instead of header and jumpers, a flat ribbon cable should be soldered to the pads for the final product. The header pads provide flexibility for the RnD process.

The design is meant to be used with the TFT_eSPI library.

Triac trigger and ZCD detecting pins are selected as IO33 and IO34. These pins are safe to use without internal confusion. Fan control pin is IO12.

A K type 800°C thermocouple with MAX6675 module is selected for temperature feedback.

Notes

All component selections are made by filtering the necessary ratings and sorting by the cheapest normally in stock component from the local supplier (özdisan).

There are just a few components which are to be sourced from other suppliers than özdisan.

- ILI9341 Touchscreen display module
- PTC air heater
- Heatsink (if not made a chassis sink)

A nextion NX3224F028 capacitive touch HMI module might be better for the UX. The programming port is usable for serial communication with the nextion module.